

Indian Institute of Technology, Bhilai

Department of Computer Science and Engineering

CSL351: Computer Networks

Assignment 11

DHCP, DNS & Web Server

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Part A: Topology Design

A.1 Network Overview

The network represents a centralized digital campus infrastructure designed to provide automatic network access, domain name resolution, and internal web hosting. It consists of a single central switch connecting 3 dedicated servers (DHCP, DNS, and Web) and 6 client end devices (PC0-PC5).

The overall structure is:

- Switch0 serves as the central node for the Local Area Network (LAN).
- DHCP Server - - - Switch0
- DNS Server - - - Switch0
- Web Server - - - Switch0
- PC0, PC1, PC2, PC3, PC4, PC5 - - - Switch0

A.2 Device Placement

The following devices were placed on the Cisco Packet Tracer workspace:

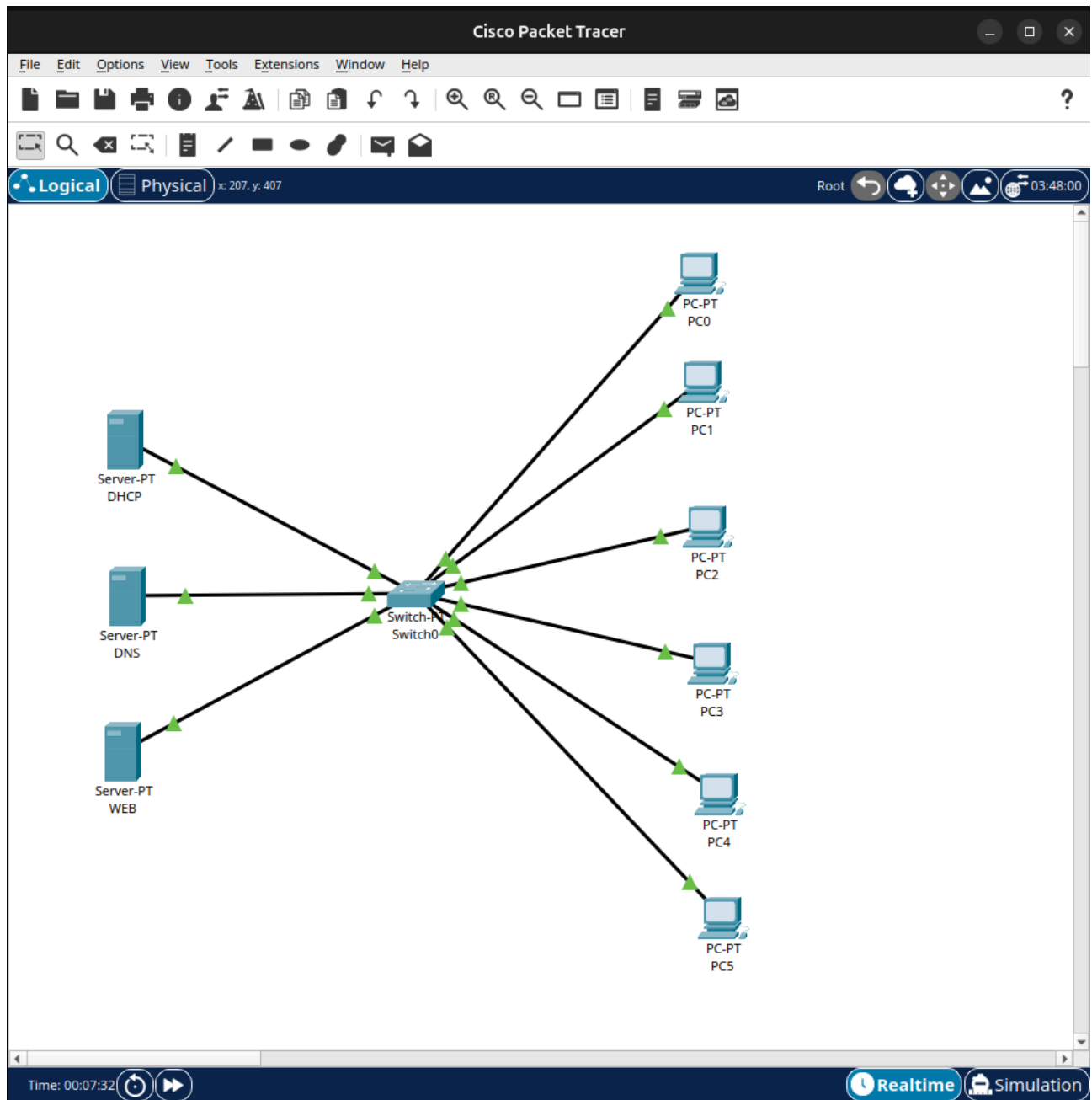
- 1 x Cisco PT Switch - Switch0
- 3 x Generic Servers - Renamed to DHCP Server, DNS Server, and WEB Server
- 6 x Generic PCs - PC0, PC1, PC2, PC3, PC4, PC5

A.3 Cable Connections

The connections were established according to the assigned network structure:

- **LAN Connections:** All 6 PCs and all 3 Servers were connected to the centralized Switch0.
- Since these are connections between different types of devices (PCs/Servers to a Switch), **Copper Straight-Through** cables were used for all links.

A.4 Network Topology Diagram



Part B: Configuration

B.1 IP Address Allocation

A single subnet 192.168.1.0/24 is used for the entire digital campus LAN. The servers are assigned static IPs, while the client PCs receive dynamic IP configurations from the DHCP server pool starting at 192.168.1.10.

Device	IP Address	Subnet Mask	Default Gateway	DNS Server
DHCP Server	192.168.1.2	255.255.255.0	192.168.1.1	192.168.1.3
DNS Server	192.168.1.3	255.255.255.0	192.168.1.1	192.168.1.3
Web Server	192.168.1.4	255.255.255.0	192.168.1.1	192.168.1.3
PC0 to PC5	Dynamic (DHCP)	255.255.255.0	192.168.1.1	192.168.1.3

B.2 Server Configuration Steps

Step 1: DHCP Server Configuration

- Click on DHCP Server → Desktop tab → IP Configuration.
- Set it to **Static**. Enter IP: 192.168.1.2, Mask: 255.255.255.0, Gateway: 192.168.1.1, DNS Server: 192.168.1.3.

- Go to the **Services** tab > **DHCP**.
- Turn the Service On.

- Set Default Gateway to 192.168.1.1 and DNS Server to 192.168.1.3.
- Set Start IP Address to 192.168.1.10.
- Click Save.

Physical

Config

Services

Desktop

Programming

Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

PRP

DHCP

Interface: FastEthernet0

Service: ☒ On ☐ Off

Pool Name: serverPool

Default Gateway: 192.168.1.1

DNS Server: 192.168.1.3

Start IP Address: 192.168.1.10

Subnet Mask: 255.255.255.0

Maximum Number of Users: 246

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add

Save

Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168.1.1	192.168.1.3	192.168.1.10	255.255.255.0	246	0.0.0.0	0.0.0.0

Step 2: DNS Server Configuration

- Click the DNS Server > Desktop tab > IP Configuration.
- Set it to Static. Enter IP: 192.168.1.3, Mask: 255.255.255.0, Gateway: 192.168.1.1, DNS Server: 192.168.1.3.

The screenshot shows the 'DNS' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is active, showing 'Static' as the selected option. The fields are filled with: IPv4 Address: 192.168.1.3, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.1.1, and DNS Server: 192.168.1.3. The 'IPv6 Configuration' section is also visible, with 'Static' selected and a Link Local Address of FE80::202:4AFF:FE81:C21B. The '802.1X' section at the bottom has 'Use 802.1X Security' unchecked, 'Authentication' set to 'MD5', and empty fields for 'Username' and 'Password'.

- Go to the Services tab > DNS.
- Turn the Service On.
- Under Resource Records, create an A Record:
- Name: campus.iitbhlai.ac.in.
- Address: 192.168.1.4 (The IP of Web Server).
- Click Add.
- Similarly add for PC0 and PC5 as well.

DNS

Physical
Config
Services
Desktop
Programming
Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

PRP

DNS

DNS Service
☒ On ☐ Off

Resource Records

Name

Type

A Record

Address

Add

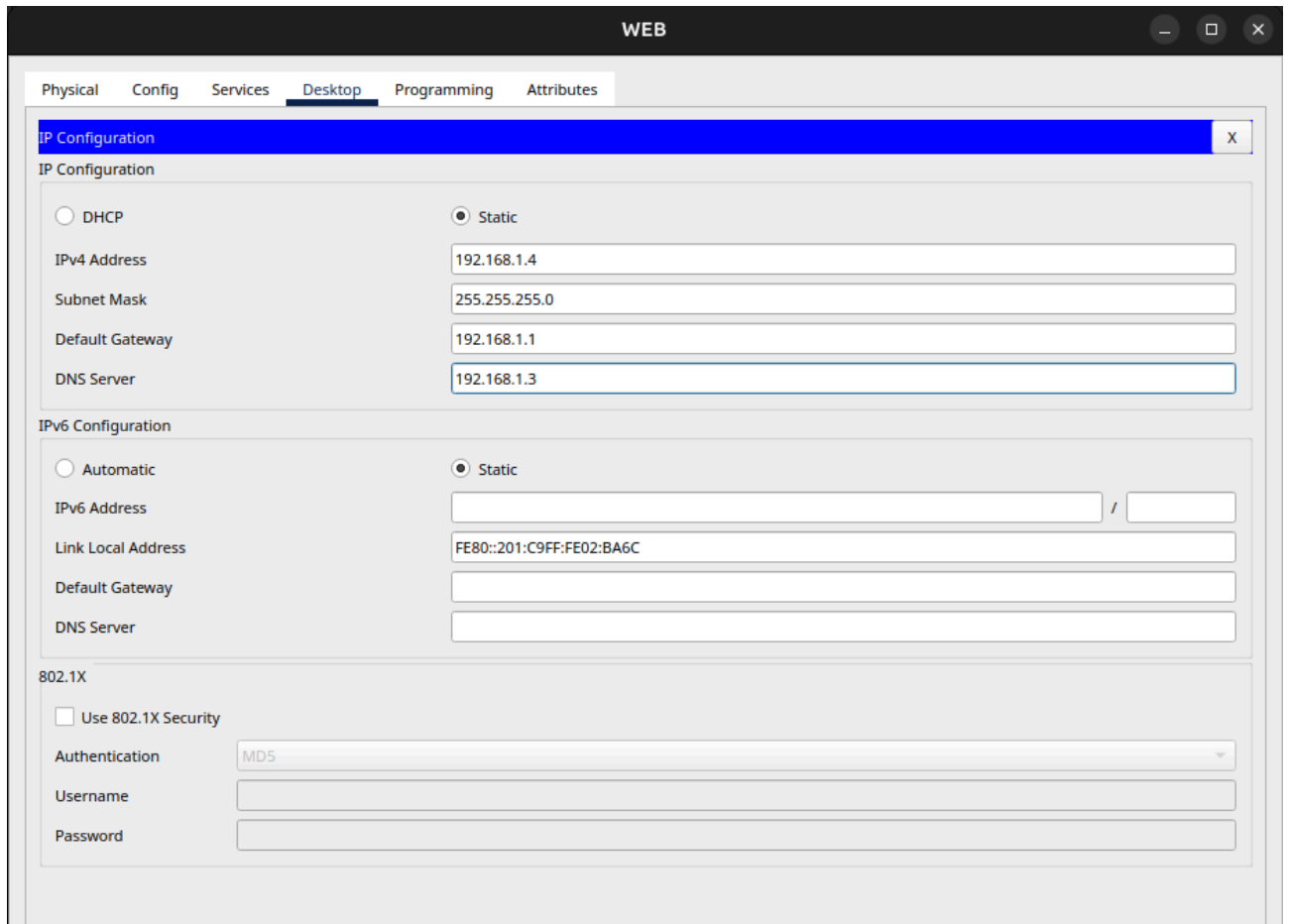
Save

Remove

No.	Name	Type	Detail
0	campus.iitbhlai.ac.in	A Record	192.168.1.4
1	pc0.campus.com	A Record	192.168.1.10
2	pc5.campus.com	A Record	192.168.1.15

Step 3: Web Server Configuration

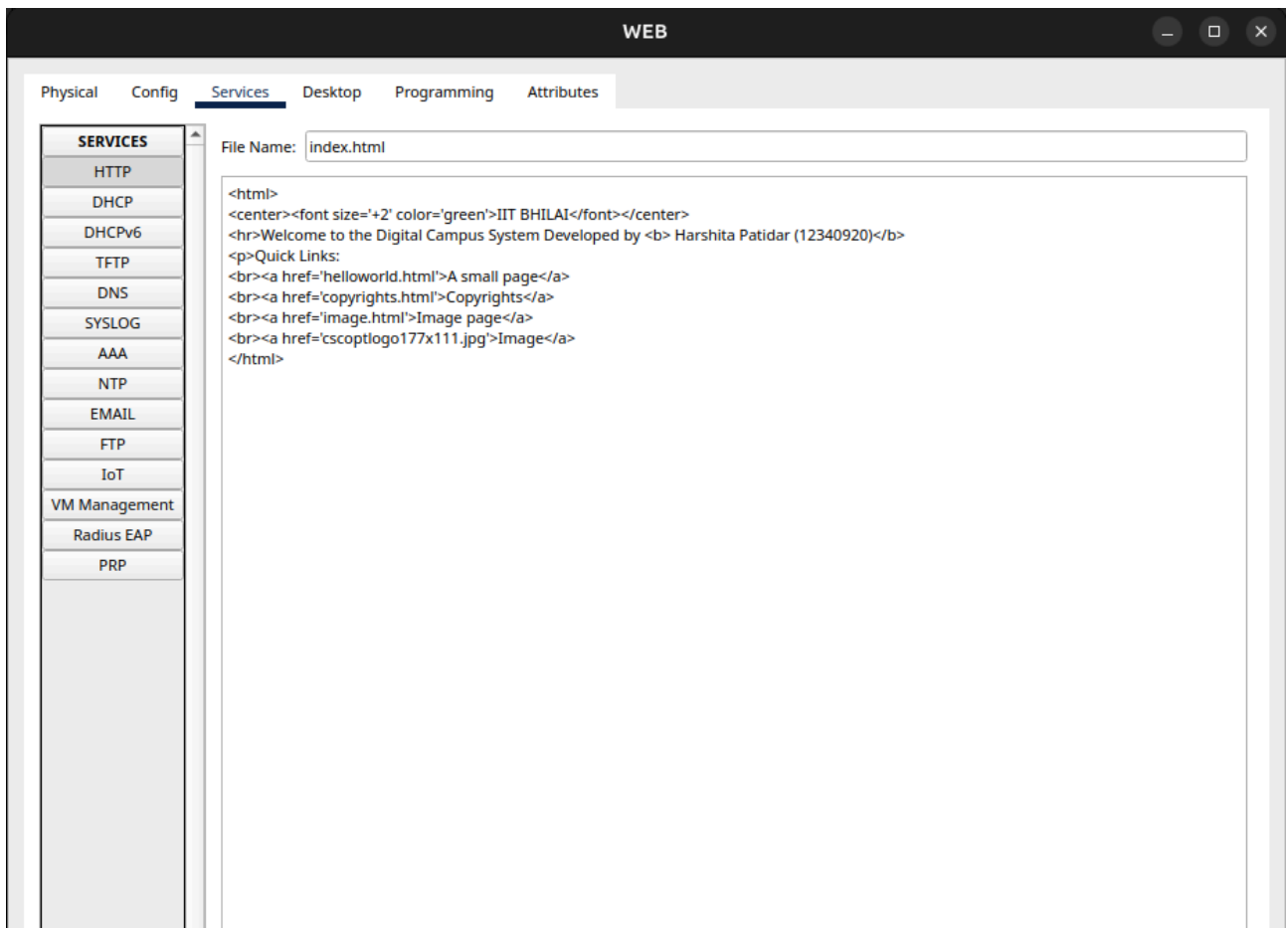
- Click the Web Server > Desktop tab > IP Configuration.
- Set it to Static. Enter IP: 192.168.1.4, Mask: 255.255.255.0, Gateway: 192.168.1.1, DNS Server: 192.168.1.3



The screenshot shows a window titled "WEB" with a tabbed interface. The "Desktop" tab is selected, and the "IP Configuration" section is active. The "Static" radio button is selected for IP Configuration. The fields are filled with: IPv4 Address: 192.168.1.4, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.1.1, and DNS Server: 192.168.1.3. The "IPv6 Configuration" section also has "Static" selected, with fields for IPv6 Address, Link Local Address (FE80::201:C9FF:FE02:BA6C), Default Gateway, and DNS Server. The "802.1X" section has "Use 802.1X Security" unchecked, and the Authentication dropdown is set to "MD5".

Section	Option	Value
IP Configuration	Mode	Static
	IPv4 Address	192.168.1.4
	Subnet Mask	255.255.255.0
	Default Gateway	192.168.1.1
	DNS Server	192.168.1.3
IPv6 Configuration	Mode	Static
	IPv6 Address	
	Link Local Address	FE80::201:C9FF:FE02:BA6C
	Default Gateway	
	DNS Server	
802.1X	Use 802.1X Security	☐
	Authentication	MD5
	Username	
	Password	

- Go to the Services tab > HTTP. Ensure HTTP and HTTPS are On.
- In the File Manager, click (edit) next to index.html.
- Change the HTML.
- Click Save.



B.3 Server Configuration Steps

To configure each PC in Cisco Packet Tracer to obtain an IP automatically via DHCP

- Click on the PC → Desktop tab → IP Configuration.
- Select the **DHCP** radio button.
- Wait for the "DHCP request successful" message to appear.

:

Below are the screenshot of all 6 PCs IP Configuration via DHCP

PC1

Physical

Config

Desktop

Programming

Attributes

IP Configuration

X

Interface

FastEthernet0

IP Configuration

☒ DHCP

☐ Static

DHCP request successful.

IPv4 Address

192.168.1.11

Subnet Mask

255.255.255.0

Default Gateway

192.168.1.1

DNS Server

192.168.1.3

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

/

Link Local Address

FE80::210:11FF:FE87:C895

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

PC3

Physical

Config

Desktop

Programming

Attributes

IP Configuration

Interface

FastEthernet0

IP Configuration

☒ DHCP

☐ Static

DHCP request successful.

IPv4 Address

192.168.1.13

Subnet Mask

255.255.255.0

Default Gateway

192.168.1.1

DNS Server

192.168.1.3

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

Link Local Address

FE80::206:2AFF:FED5:B7CA

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

PC4

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP
 ☐ Static

DHCP request successful.

IPv4 Address

192.168.1.14

Subnet Mask

255.255.255.0

Default Gateway

192.168.1.1

DNS Server

192.168.1.3

IPv6 Configuration

☐ Automatic
 ☒ Static

IPv6 Address

 /

Link Local Address

FE80::209:7CFF:FE05:4027

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

PC5

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP
 ☐ Static

DHCP request successful.

IPv4 Address

192.168.1.15

Subnet Mask

255.255.255.0

Default Gateway

192.168.1.1

DNS Server

192.168.1.3

IPv6 Configuration

☐ Automatic
 ☒ Static

IPv6 Address

 /

Link Local Address

FE80::201:64FF:FE75:87AB

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

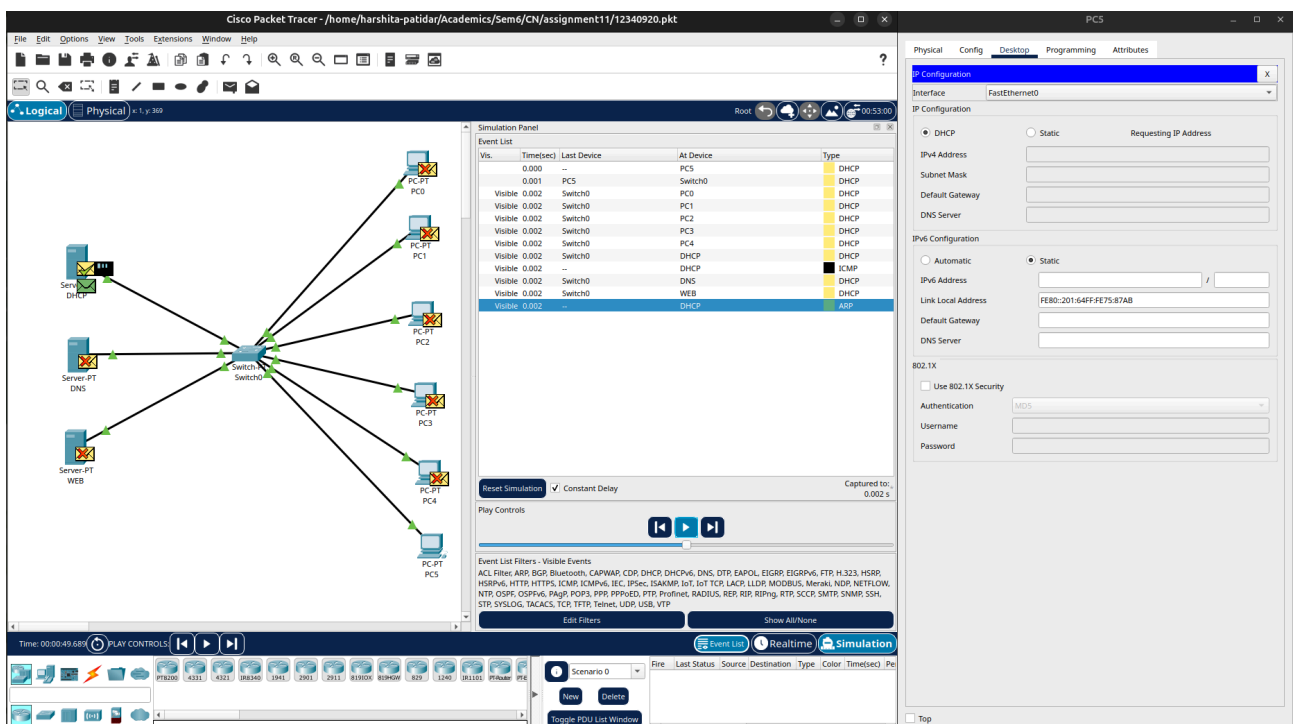
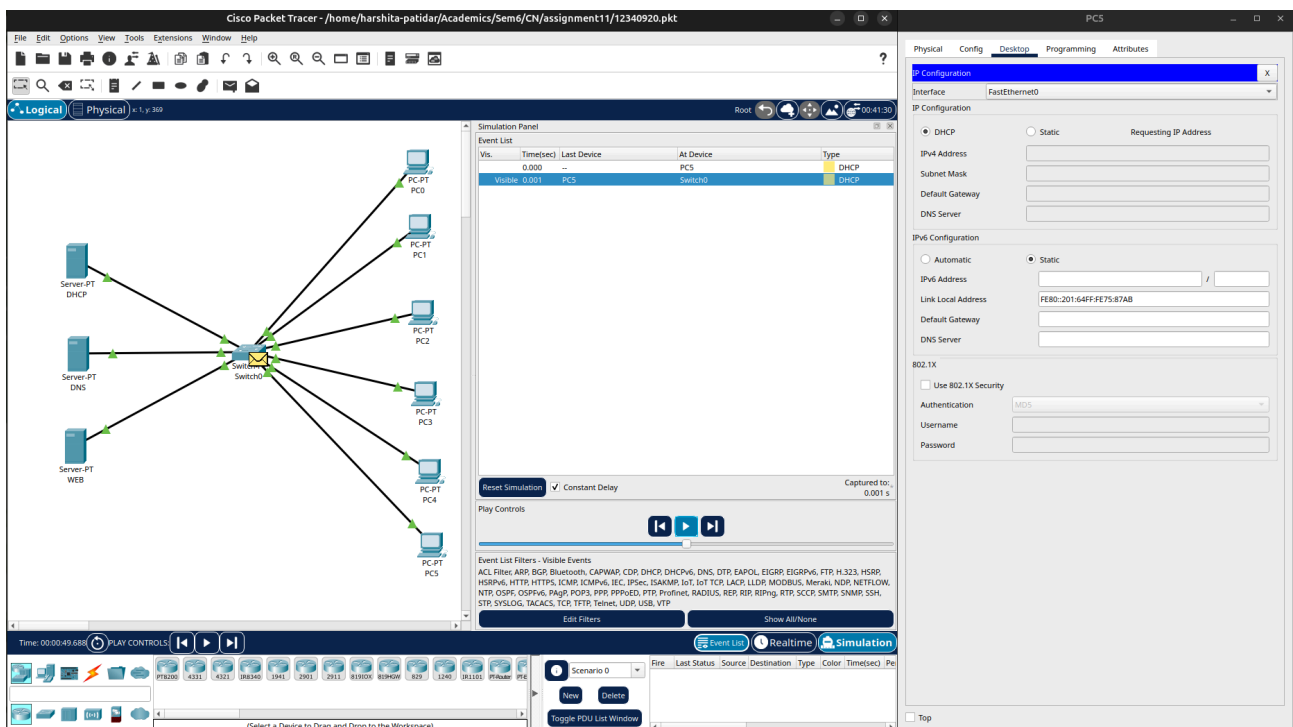
Part C: Verification

C.1 Dynamic IP Allocation Verification (DHCP)

To verify that the DHCP server correctly allocated addresses, the IP Configuration window of each client was checked. All PCs successfully received dynamic IP configurations, including the correct Subnet Mask, Default Gateway, and DNS Server.

(same screenshot as above section B.3 showing dynamic IP Allocation verification)

Also please find below the screenshots taken in simulation mode to demonstrate IP Allocation via DHCP for PC5



Cisco Packet Tracer - /home/harshita-patidar/Academics/Sem6/CN/assignment11/12340920.pkt

Event List

Vis.	Time(sec)	Last Device	At Device	Type
0.000	—	PC5	Switch0	DHCP
0.001	PC5	Switch0	PC5	DHCP
0.002	Switch0	PC5	DHCP	DHCP
0.002	Switch0	PC2	DHCP	DHCP
0.002	Switch0	PC3	DHCP	DHCP
0.002	Switch0	PC4	DHCP	DHCP
0.002	Switch0	DHCP	DHCP	DHCP
0.002	Switch0	DNS	DHCP	DHCP
0.002	Switch0	WEB	DHCP	DHCP
0.002	Switch0	DHCP	DHCP	DHCP
Visible: 0.003	DHCP	Switch0	ARP	ARP

Reset Simulation Constant Delay Captured to: 0.003 s

IP Configuration

Interface: FastEthernet0

IP Configuration

- ☒ DHCP ☐ Static Requesting IP Address
- IPv4 Address:
- Subnet Mask:
- Default Gateway:
- DNS Server:

IPv6 Configuration

- ☐ Automatic ☒ Static
- IPv6 Address:
- Link Local Address: FE80::201:64FF:FE75:87AB
- Default Gateway:
- DNS Server:

802.1X

- ☐ Use 802.1X Security
- Authentication: MD5
- Username:
- Password:

Time: 00:00:49.000 PLAY CONTROLS

Cisco Packet Tracer - /home/harshita-patidar/Academics/Sem6/CN/assignment11/12340920.pkt

Event List

Vis.	Time(sec)	Last Device	At Device	Type
0.004	Switch0	PC5	ARP	ARP
0.004	Switch0	PC4	ARP	ARP
0.004	Switch0	DNS	ARP	ARP
0.004	Switch0	WEB	ARP	ARP
0.920	—	Switch0	STP	STP
0.921	Switch0	PC2	STP	STP
0.921	Switch0	PC0	STP	STP
0.921	Switch0	PC3	STP	STP
0.921	Switch0	PC1	STP	STP
0.921	Switch0	PC5	STP	STP
0.921	Switch0	PC4	STP	STP
0.921	Switch0	DHCP	STP	STP
0.921	Switch0	DNS	STP	STP
0.921	Switch0	WEB	STP	STP
1.002	—	DHCP	ICMP	ICMP
1.002	—	DHCP	ARP	ARP
1.507	DHCP	Switch0	DHCP	DHCP
Visible: 1.508	Switch0	PC0	DHCP	DHCP
Visible: 1.508	Switch0	PC1	DHCP	DHCP
Visible: 1.508	Switch0	PC2	DHCP	DHCP
Visible: 1.508	Switch0	PC3	DHCP	DHCP
Visible: 1.508	Switch0	PC4	DHCP	DHCP
Visible: 1.508	Switch0	DNS	DHCP	DHCP
Visible: 1.508	Switch0	WEB	DHCP	DHCP
Visible: 1.508	Switch0	DHCP	DHCP	DHCP

Reset Simulation Constant Delay Captured to: 1.508 s

IP Configuration

Interface: FastEthernet0

IP Configuration

- ☒ DHCP ☐ Static Requesting IP Address
- IPv4 Address:
- Subnet Mask:
- Default Gateway:
- DNS Server:

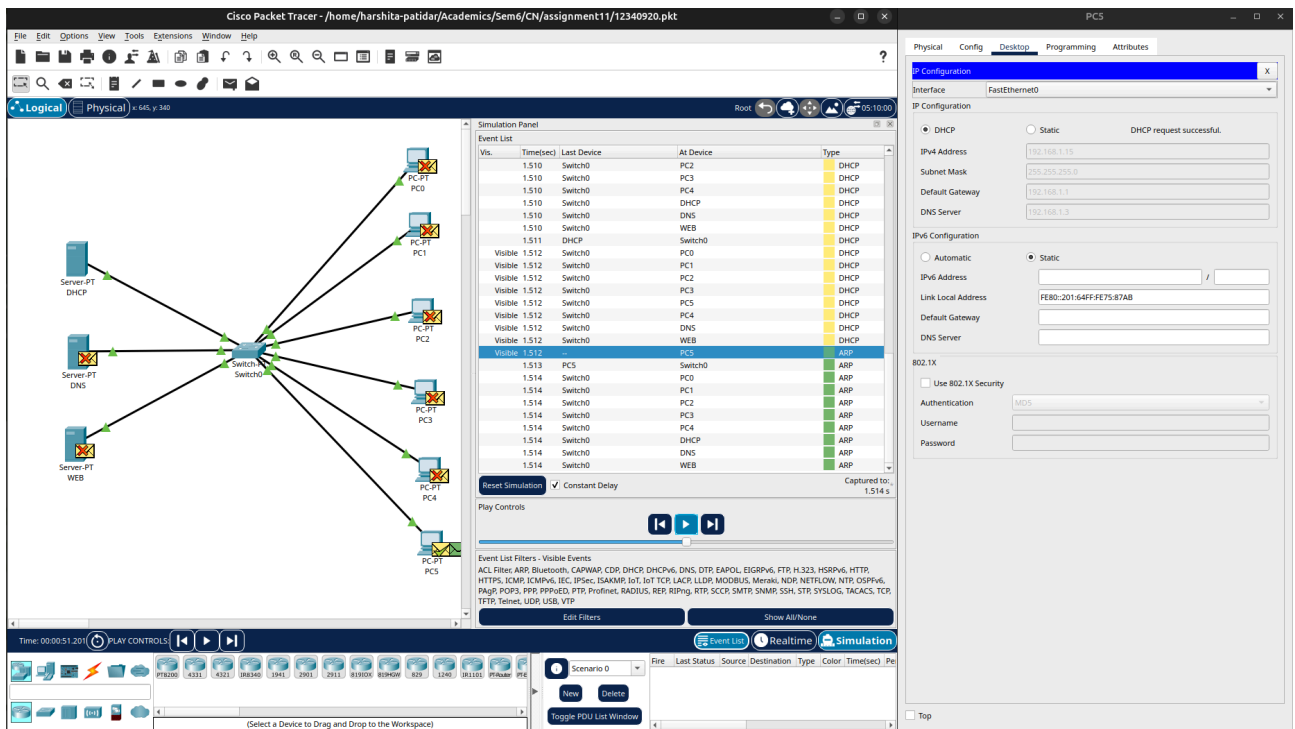
IPv6 Configuration

- ☐ Automatic ☒ Static
- IPv6 Address:
- Link Local Address: FE80::201:64FF:FE75:87AB
- Default Gateway:
- DNS Server:

802.1X

- ☐ Use 802.1X Security
- Authentication: MD5
- Username:
- Password:

Time: 00:00:51.195 PLAY CONTROLS



(DHCP Request successful, PC5 IP configuration done via DHCP, for other PCs please find attached screenshots in the previous section)

C.2 Ping Tests

To test seamless communication and resolution, the Command Prompt of a sourcePC (PC5) was used to ping another client (PC0) and PC (PC0) was used to ping client (PC5).

Test 1: Ping via IP Address PC0 successfully pinged PC5 using its dynamically assigned IP address (192.168.1.15).

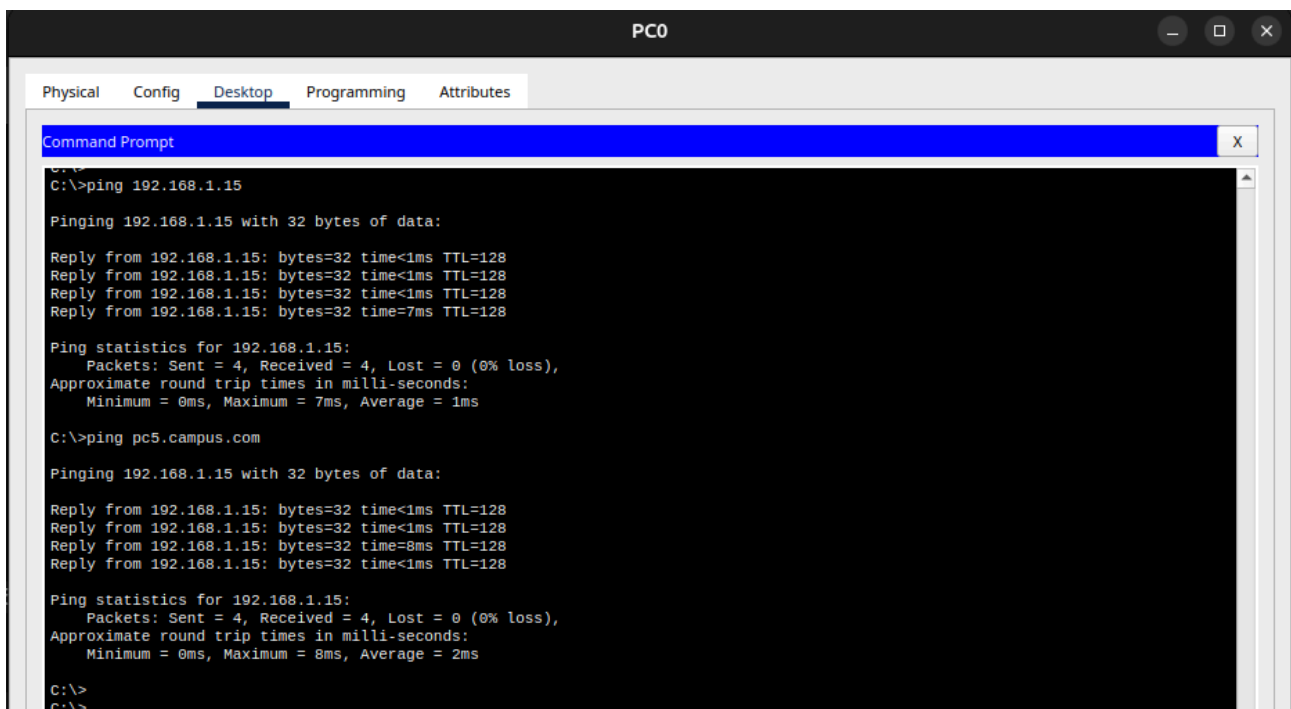
```
C:\> ping 192.168.1.15
```

Result: 4 replies received, 0% packet loss.

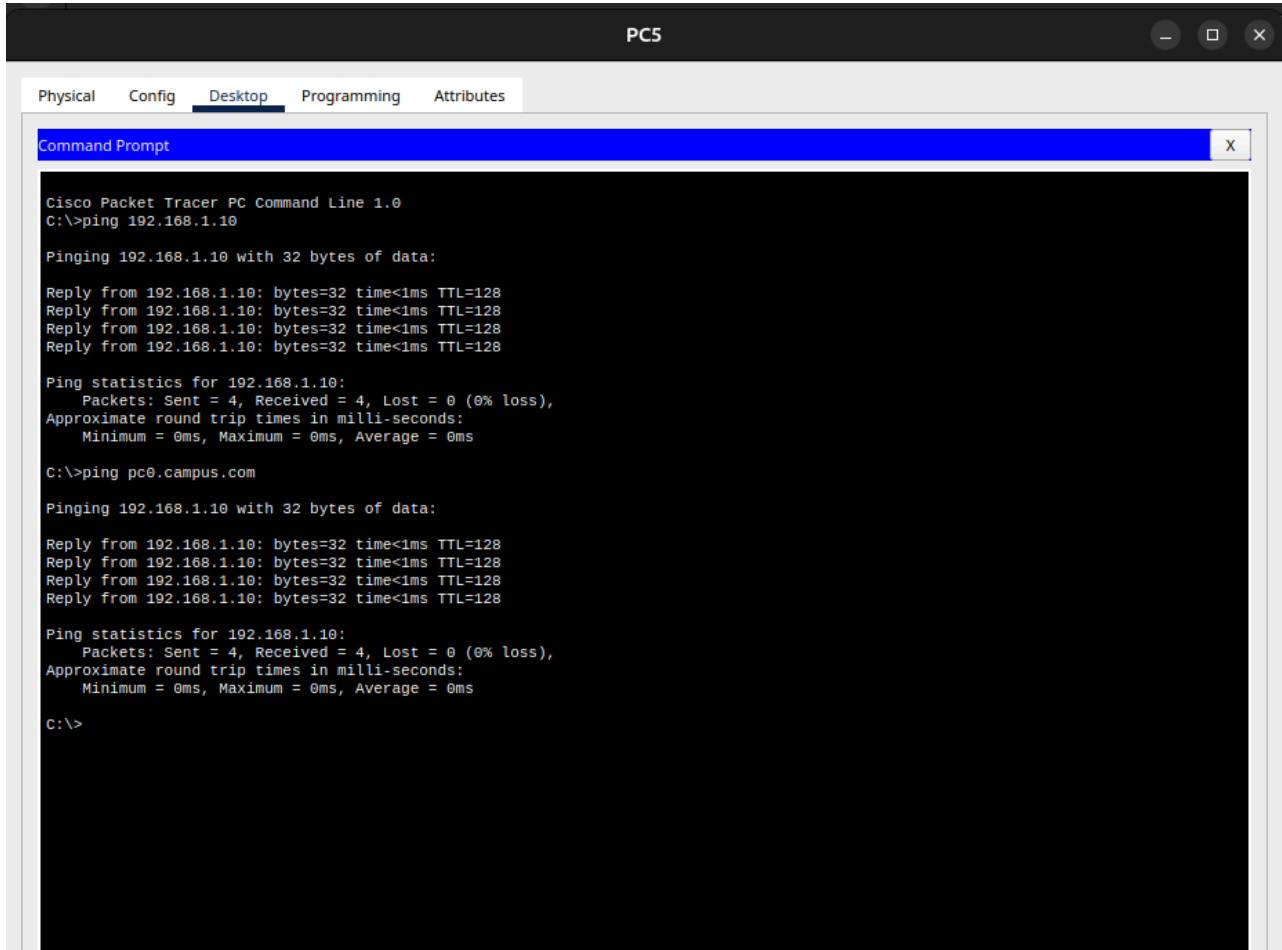
Test 2: Ping via Domain Name Utilizing the A records established in the DNS server, PC0 successfully pinged PC5 using its assigned domain name (pc5.campus.com)

```
C:\> ping pc5.campus.com
```

Result: The DNS server successfully resolved the name to 192.168.1.15, resulting in 4 replies with 0% packet loss.



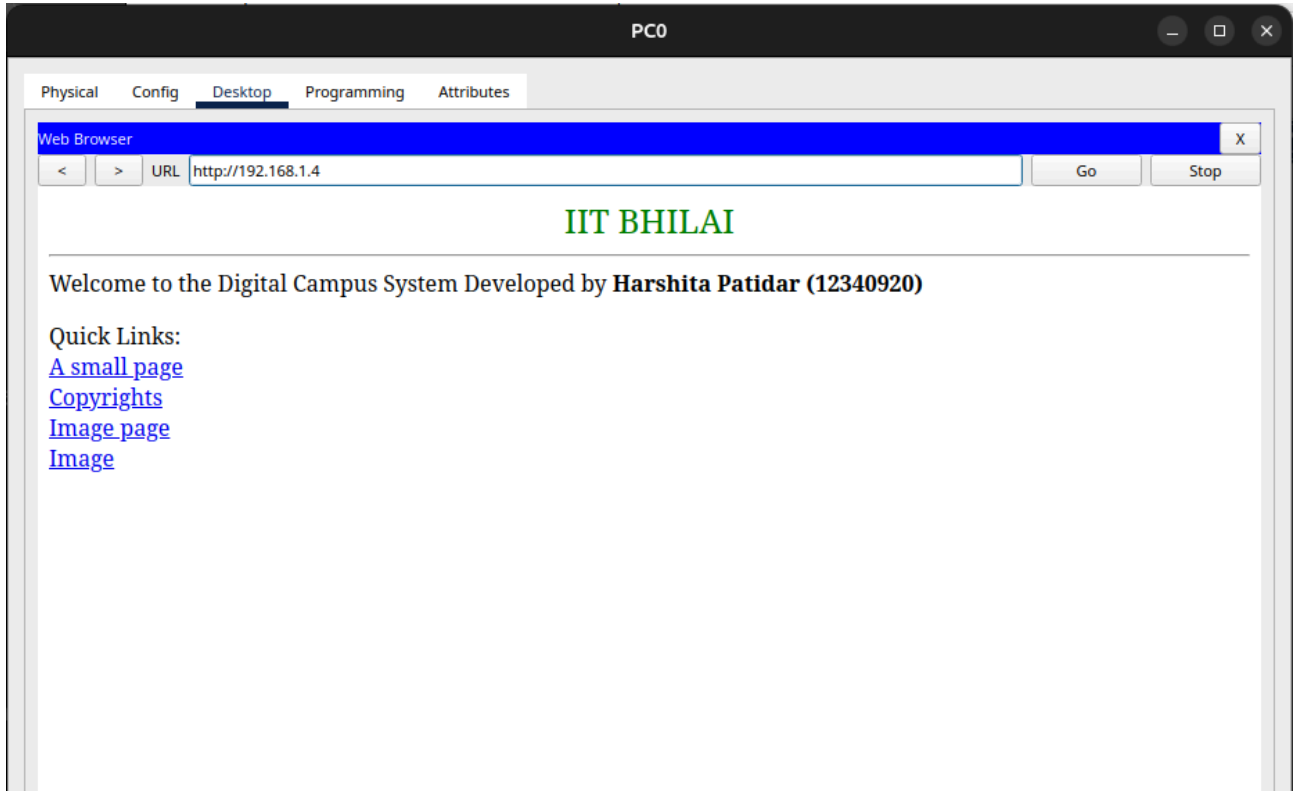
Similarly, same was repeated using PC5 command prompt to ping PC0 via both methods.



C.3 Web Server Access

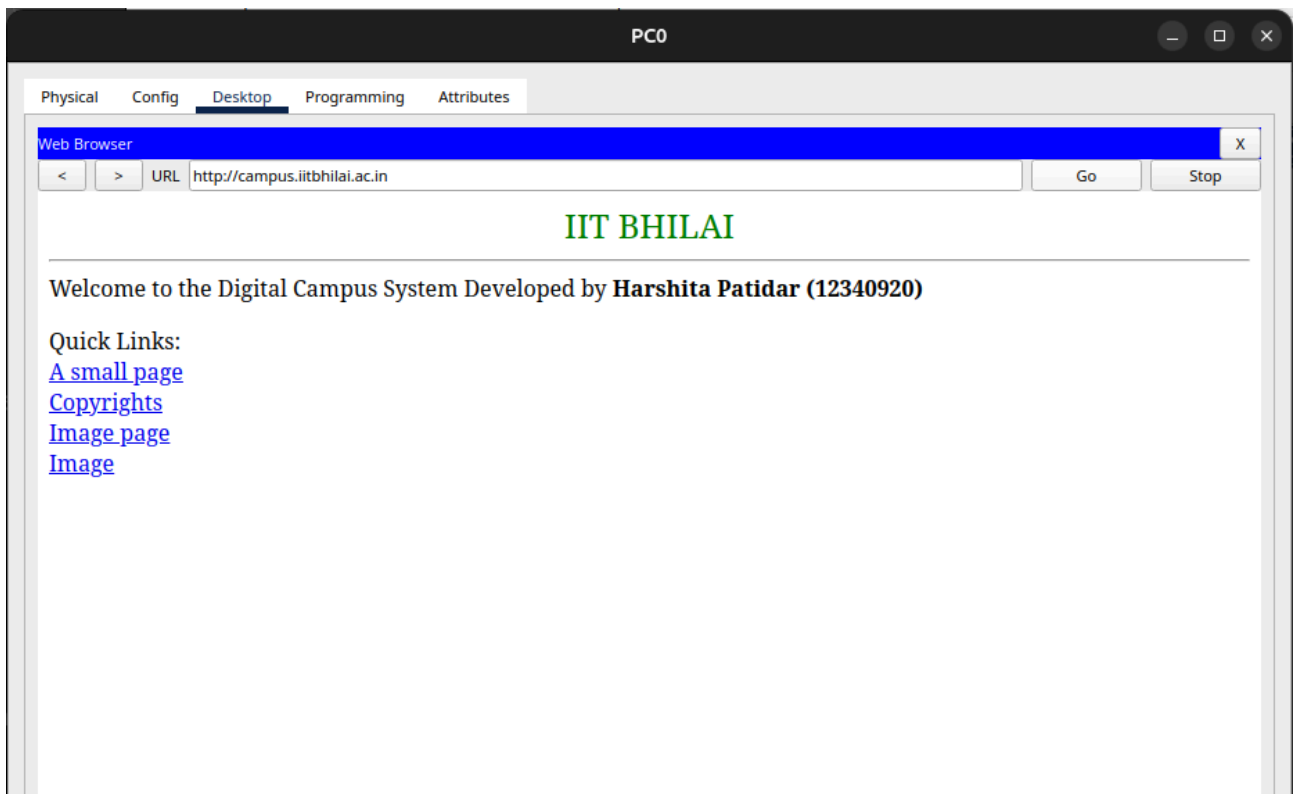
Client access to the internal web server was verified using the built-in Web Browser utility on PC0.

Test 1: Accessing via IP Address The customized HTML page loaded successfully when navigating to the explicit IP. **URL:** `http://192.168.1.4`



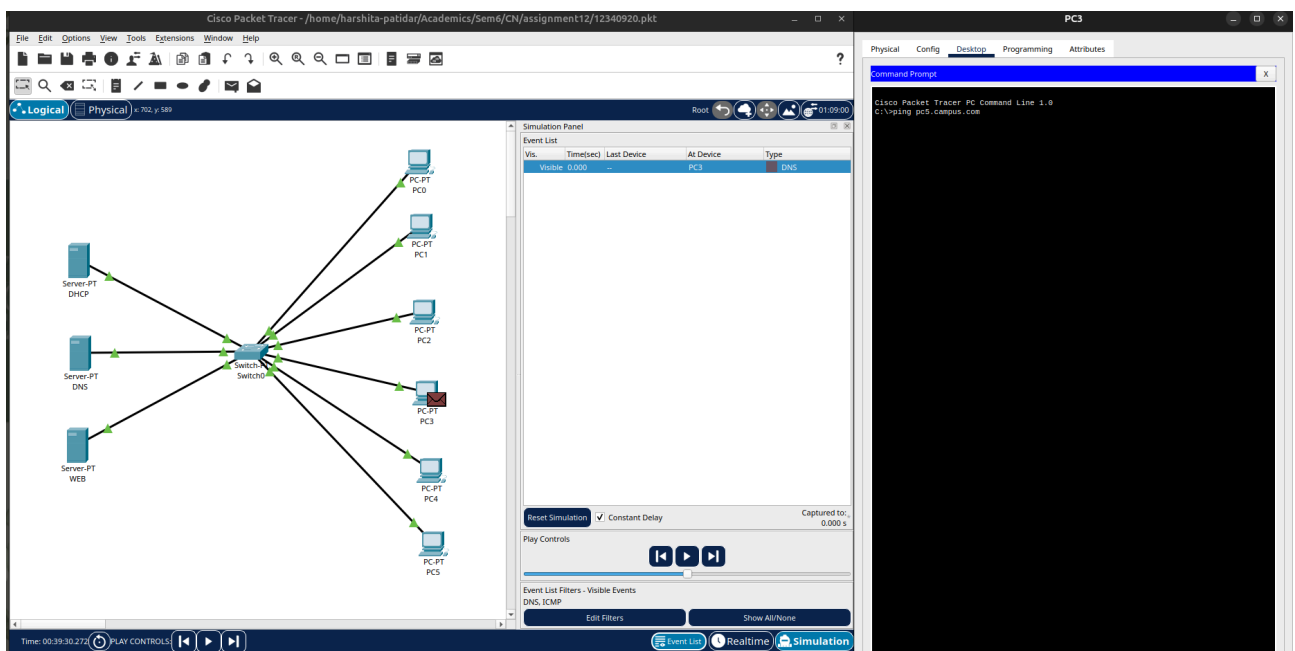
Test 2: Accessing via Domain Name The customized HTML page loaded successfully when navigating to the campus domain.

URL: `http://campus.iitbhilai.ac.in`



C.4 DNS Resolution Verification

To explicitly visualize and verify the DNS resolution process, Cisco Packet Tracer's Simulation Mode was utilized. A network request was initiated from PC3 using a domain name instead of a direct IP address.

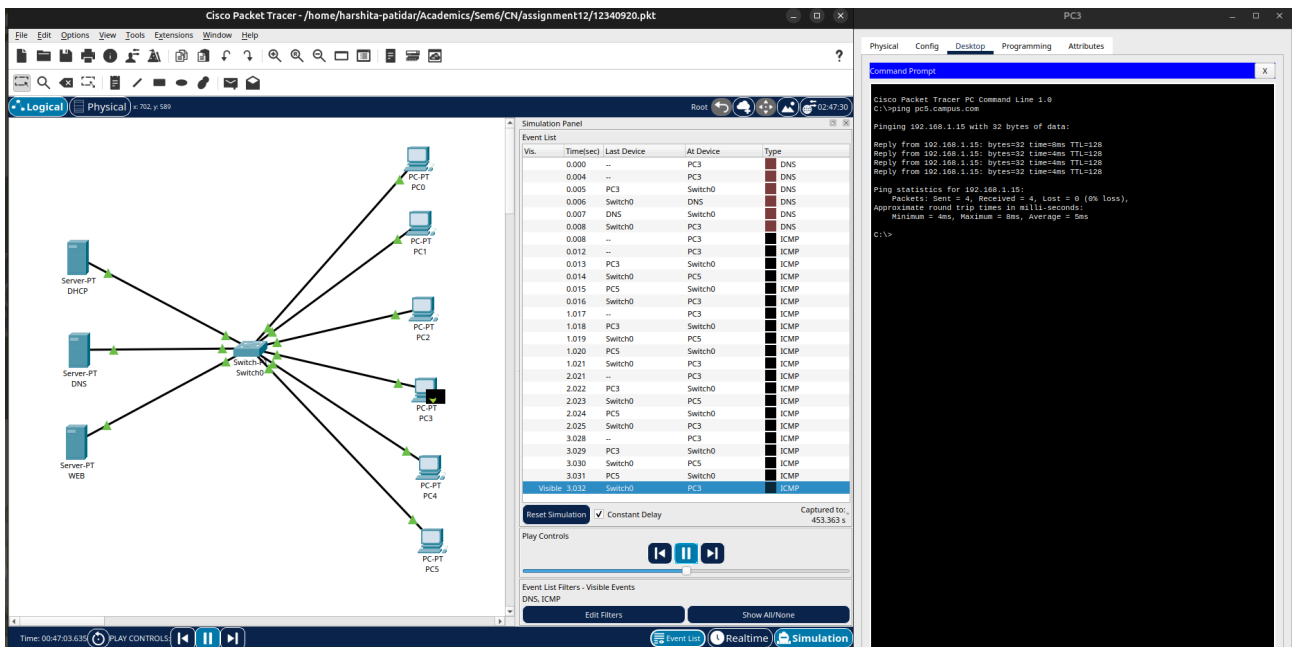
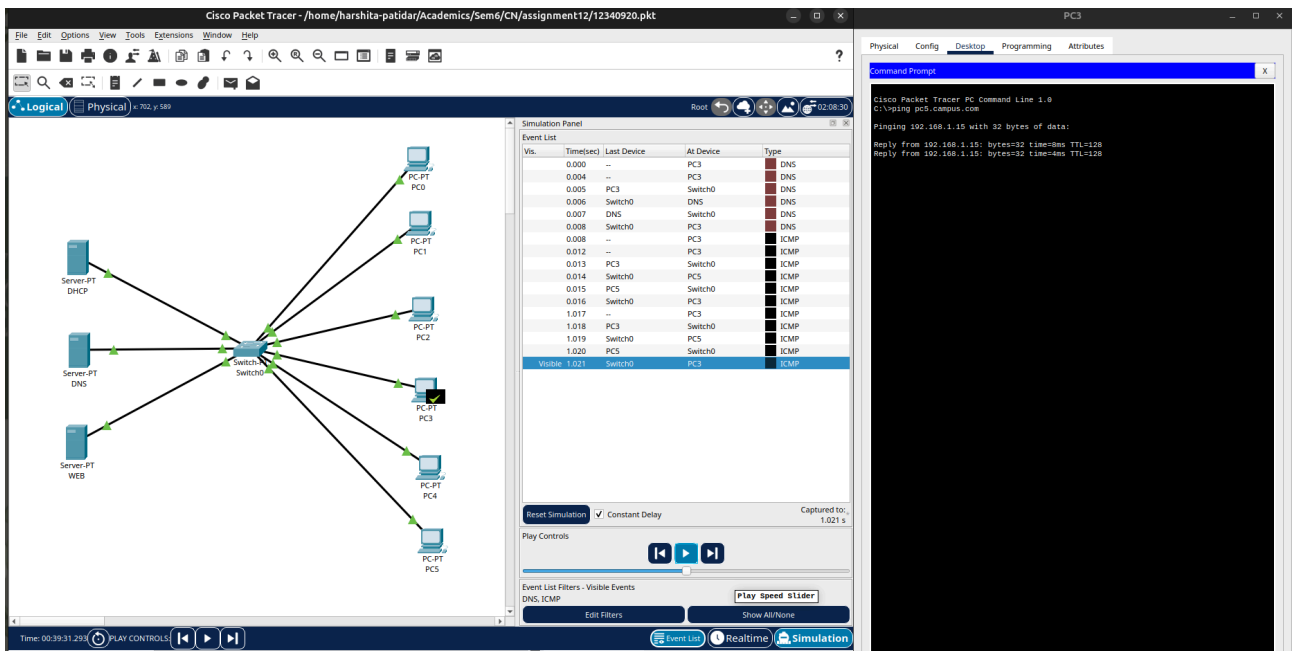


The network diagram shows a central Switch0 connected to three servers (DHCP, DNS, WEB) and five PCs (PC0-PC4). The PC3 command prompt shows the command `C:\>ping pc3.campus.com`.

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC3	DNS
	0.004	--	PC3	DNS
	0.005	PC3	Switch0	DNS
Visible	0.006	Switch0	DNS	DNS

The network diagram is the same as the first image. The PC3 command prompt shows the command `C:\>ping 192.168.1.15 with 32 bytes of data:`.

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC3	DNS
	0.004	--	PC3	DNS
	0.005	PC3	Switch0	DNS
	0.006	Switch0	DNS	DNS
	0.007	DNS	Switch0	DNS
Visible	0.008	Switch0	PC3	DNS
Visible	0.008	--	PC3	ICMP



Analysis of Packet Flow: As illustrated in the simulation capture above, the system does not immediately send an ICMP (ping) packet. Instead, PC3 first constructs a DNS Query packet. This packet is routed through the switch directly to the configured DNS Server (192.168.1.3) on Port 53. The DNS Server receives this inbound Protocol Data Unit (PDU), processes the requested name against its A Records, and returns the mapped IPv4 address back to PC3. Only after this successful resolution does PC3 construct and transmit the actual ICMP echo requests. This packet-level verification conclusively demonstrates that the DNS service is actively intercepting and correctly resolving domain names across the digital campus network.